

Hide menu

- Lesson 5. Types of Attacks
- Lesson 6. Frequency Analysis of Monoalphabetic Ciphers - Single-Character Frequencies
- Lesson 7. Frequency Analysis of Monoalphabetic Ciphers - Multi-Character Frequencies
- Lesson 8. Frequency Analysis for Monoalphabetic Ciphers - Example
- Lesson 9. Key Length Determination in Polyalphabetic Ciphers
- Lesson 10. Example of Cracking a Vigenere Cipher
- Module 3 Discussion
- Module Assessment

Quiz: Graded Assessment - Cryptanalysis
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Graded Assessment - Cryptanalysis

Review Learning Objectives

1. Attacks that exploit weaknesses in the implementation of a cryptographic system are generally known as 1 / 1 point

- ☐ social engineering attacks.
- ☐ timing attacks.
- ☐ Related key attacks.
- ☒ side channel attacks.

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While there are many types of such attacks, this is the name given to them collectively.

2. The term often used to describe using force or threats to collect information from a cryptosystem is 1 / 1 point

- ☒ rubber-hose cryptanalysis.
- ☐ social engineering.
- ☐ traffic analysis.
- ☐ extortion.

Correct
This tongue-in-cheek name is a catch-all term for such methods.

3. Frequency analysis exploits what characteristic of monoalphabetic ciphers? 1 / 1 point

- ☒ The underlying frequency of the plaintext languages.
- ☐ The frequency distribution of word lengths.
- ☐ The frequency of vowels at the beginning of words.
- ☐ The number of distinct symbols in the cipher text.

Correct
A monoalphabetic cipher does not hide the plaintext language statistics.

4. Which of the following does not typically complicate the use of frequency analysis. 1 / 1 point

- ☐ Limiting messages to fairly short lengths.
- ☐ Using infrequently used letters as common punctuation marks.
- ☐ Using short abbreviations for longer words or concepts.
- ☒ Spelling out numbers and other non-text items.

Correct
This has very little impact on the frequency distribution and increases the amount of ciphertext available to analyze.

5. In the English alphabet, how many digraphs are there? 1 / 1 point

- ☐ 17,576
- ☒ 676
- ☐ 456,976
- ☐ 26

Correct
This is 26 squared.

6. A principle motivation in developing polyalphabetic ciphers was 1 / 1 point

- ☒ to flatten the frequency distribution of the ciphertext characters.
- ☐ to overcome the fact that nearly all monoalphabetic ciphers had already been used.
- ☐ to reduce the size of the key to something that could be easily remembered.
- ☐ to increase the size of the key space.

Correct
If the ciphertext characters appear uniformly and seemingly at random, frequency analysis is rendered very ineffective.

7. What is the index of coincidence for a language that uses it's alphabet randomly. 1 / 1 point

- ☒ 1.0
- ☐ 1.73
- ☐ 1/N, where N is the number of characters in the alphabet.
- ☐ 0.0

Correct
The index of coincidence is the ratio of the actual rate of coincidence divided by the rate for a randomly used alphabet.

8. A Vigenere cipher using a particular non-English alphabet has a keylength of 20. We would like the most common plaintext letter (which has a frequency of about 10%) to appear about 5 times in each key-letter group. About how much ciphertext do we need access to? 0 / 1 point

- ☒ 50
- ☐ 20
- ☐ Indeterminate without knowing the length of the alphabet.
- ☐ 1000 characters.

Incorrect
This means that the most frequent letter will appear about 5 times in the entire ciphertext, but it will be spread out among 20 groups.

9. The coincidence rate of a Vigenere cipher peaks strongly at offset distances of 35 and 42. Key lengths to focus on quickly include 1 / 1 point

- ☒ 7
- ☐ 5, 6, or 7
- ☐ 35 or 42
- ☐ Indeterminate, since 42 is not a multiple of 35.

Correct
It is likely that 35 and 42 are both multiples of the key length.

10. The fact that some letters are almost always (or almost never) followed by certain other letters can help with all of the following except 1 / 1 point

- ☒ helping to "break ties" when single-character frequency analysis fails to distinguish mappings.
- ☐ Calculating the correct single-character frequencies in the ciphertext.
- ☐ Help to identify ciphertext characters that probably represent vowels.
- ☐ helping to identify potential breaks between words.

Correct
Single characters that have about the same frequency by themselves have significantly different frequencies relative to the letters next to them.